



Cambridge Lower Secondary Checkpoint

SCIENCE

0893/01

Paper 1

April 2025

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Markers were instructed to award marks. It does not indicate the details of the discussions that took place at a Markers' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the End of Series Report. Cambridge will not enter into discussions about these mark schemes.

This document has **16** pages.

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed so answers do not need to be grammatically correct. Significant figures will be indicated in the question or in the mark scheme. Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>Underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working, provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

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Question	Answer	Marks	Further Information
1(a)	absorbed by <u>root hair</u> (cells)	2	each correct answer = 1 mark Accept enters by osmosis or diffusion Note just root is not sufficient Ignore reference to xylem
1(b)	mineral(s) (salts)	1	Accept named mineral salts, e.g. nitrates, magnesium (ions or salts) Ignore nutrients / nitrogen / vitamins Do not accept proteins
1(c)	leaves transpiration	1	Both correct for the mark Accept stomata or stoma Ignore pores Accept evaporation or diffusion or evapotranspiration

Question	Answer	Marks	Further Information
2(a)	(conical) flask	1	Accept (Erlenmeyer) flask Ignore round bottom or measuring or glass
2(b)(i)	mass is conserved in chemical reactions	1	Accept mass is not gained or lost in a chemical reaction Accept (because of) Law of conservation of mass Accept number of atoms stays the same, they are just rearranged
2(b)(ii)	gas produced	1	Accept carbon dioxide produced Ignore just name of incorrect gas but accept incorrect gas if qualified by the word gas Ignore reference to evaporation
2(c)	rate decreases	1	Note the answer must be comparative Accept the reaction is slower but reaction is slow is not sufficient Ignore any explanation

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Question	Answer	Marks	Further Information
3	any three from M1 water changes (from liquid) to gas M2 (idea that) evaporation absorbs energy M3 (thermal) energy is transferred from the hand to the water M4 particles with higher energy escape M5 leave behind particles with lower energy	3	each correct answer = 1 mark Accept (liquid) water changes into vapour Accept endothermic or evaporation requires energy Accept as water particles evaporate, it decreases the amount of (thermal) energy on Mike's hand Accept heat for thermal energy Accept particles with higher speed escape Accept leave behind particles with slower speed Accept breaking the attraction between particles requires energy = 2 marks

Question	Answer	Marks	Further Information
4(a)(i)	attraction between a positive ion and a negative ion or electrostatic attraction between ions	1	Note answer must refer to ions and not just charges Accept bonding between a positive and negative ion Accept cation for positive ion and anion for negative ion Do not accept references to sharing electrons or molecules
4(a)(ii)	magnesium (atom) loses two electrons or gives away two electrons (to make a magnesium ion) oxygen (atom) gains two electrons (to make an oxide ion)	2	each correct answer = 1 mark Accept two electrons move from magnesium (atom) to oxygen (atom) = 2 marks Note if no other mark awarded accept electron(s) move from magnesium (atom) to oxygen (atom) or magnesium loses electrons and oxygen gains electrons = 1 mark

Question	Answer	Marks	Further Information
4(b)	both iodide ions are negatively charged	1	Accept both iodide ions have the same charge Accept iodide ions repel each other Accept iodine ions for iodide ions Accept ionic bonds (only) form between positive and negative ions Accept cation for positive ion and anion for negative ion

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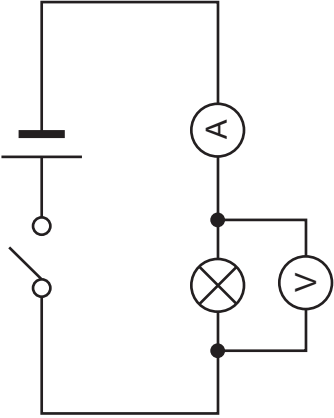
Question	Answer	Marks	Further Information
5(a)	part of the model biscuit syrup candle part of the Earth mantle core crust	2	all three correct = 2 marks one or two correct = 1 mark more than one line drawn from part of model to two or more parts of Earth = 0 marks for that part of model
5(b)	(safety risk) hot syrup (could splash into eyes) (how Oliver controls this risk) wears (safety) goggles or (safety risk) equipment is hot or syrup is hot / hands can be burnt (how Oliver controls this risk) wears (heat proof or safety) gloves or (safety risk) syrup is hot (could spill onto body) (how Oliver controls this risk) stand up / wears apron / wears laboratory coat	2	correct safety risk or correct control of a risk = 1 mark correct safety risk and linked control = 2 marks Accept burns Note wearing safety gear is not sufficient Ignore move away from the equipment Ignore candle flame and reduce the size of the candle flame

Question	Answer	Marks	Further Information
5(c)	M1 (description) biscuits move apart and any two from (explanation) M2 (idea that) movement of syrup pulls the biscuit M3 convection current produced M4 hot syrup moves upwards (above the candle) / hot syrup is less dense M5 cold syrup moves downwards (at the edges of the container) / cold syrup is more dense	3	each correct answer = 1 mark Accept marking points shown on a diagram Accept tectonic plates for biscuit Accept mantle for syrup Accept core for candle Note biscuits move is not sufficient Note this may be shown by arrows on the biscuits Accept hot syrup less dense than cold syrup Ignore hot particles rise or hot particles are less dense Ignore references to heated particles Ignore cold particles sink or cold particles are more dense Ignore references to cooled particles

Question	Answer	Marks	Further Information
6(a)	chloroplast	1	Ignore chlorophyll
6(b)	carbon dioxide + water → (oxygen) + glucose	2	reactants either order = 1 mark glucose as product = 1 mark Ignore sugar and carbohydrate Accept correct formulae, CO₂, H₂O and C₆H₁₂O₆, but names take precedence Ignore incorrect balancing
6(c)(i)	4	1	Accept any indication of the correct answer, e.g. underlining or ticking, but circling in the table takes precedence Accept rate 24 or light intensity 3
6(c)(ii)	36	1	Accept answer written next to question, but answer in table takes precedence

Question	Answer	Marks	Further Information
6(c)(iii)	correct label including unit and linear scale for y-axis all points plotted correctly straight line of best fit excluding the anomalous point	3	each correct answer = 1 mark Accept rate (of oxygen produced) in cm^3/hour and linear scale that ensures that the points occupy half of the graph paper Accept tolerance $\pm$ half a small square Accept ecf from incorrect or missing calculation Accept if volume of gas plotted as the y-axis Accept ecf from incorrect plotting or from volume of gas as the y-axis Ignore any part of line drawn between light intensity 0 and 1 Note must be drawn with a ruler Do not accept dot-to-dot

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Question	Answer	Marks	Further Information
7(a)(i)		1	Accept voltmeter symbol drawn connected either side of the lamp in parallel Ignore missing junction of conductor symbols Note one voltmeter connection must be between left of cell and lamp and the other connection between the right of cell and lamp Do not accept voltmeter symbol with a line through it or voltmeter in series
7(a)(ii)	4.5 (A)	1	Accept 4.4 or 4.6
7(a)(iii)	(idea that) the lamps are less bright	1	Accept higher level answers, e.g. the current (in the circuit) is reduced Accept amps decrease or resistance increases Accept power decreases Ignore current slows down Note voltage decreases is not sufficient but accept voltage across each lamp is decreased Do not accept voltage increases Ignore other comments about voltage
7(b)	$A_2 = 3 \text{ (A)}$ $A_3 = 3 \text{ (A)}$	1	Both correct for the mark

Question	Answer	Marks	Further Information
7(c)	$\text{(resistance =) } \frac{\text{voltage}}{\text{current}} \text{ or } \frac{V}{I} \text{ or } \frac{12.0}{2.5}$ (resistance =) 4.8 (unit) Ω or ohm(s)	3	Accept 4.8 Ω with or without working out = 3 marks Accept 4.8 with or without working out = 2 marks correct equation or evidence of correct substitution into equation = 1 mark Note if both equation and substitution given, the equation takes precedence correct unit = 1 mark

Question	Answer	Marks	Further Information
8	(A) combustion (F) feeding	2	each correct answer = 1 Accept burning Accept feed or eating Ignore nutrition or ingestion or consumption

Question	Answer	Marks	Further Information
9(a)	no more left alive	1	Accept zero population or species completely eradicated or no more left (on Earth) or species no longer exists
9(b)	any two from change in environmental conditions (idea of new or extra) predators (idea of new) disease lack of habitat or change in habitat hunters lack of food	2	each correct answer = 1 mark Accept examples of environmental changes e.g. global warming, rising sea levels, volcanic eruption, climate change, pollution Accept examples, e.g. deforestation Accept (over) hunting Accept introduction of invasive species or (increased) competition Accept death rate is greater than reproduction rate / unable to produce enough offspring / no reproduction

Question	Answer	Marks	Further Information
10	oxygen or water	1	

Question	Answer	Marks	Further Information
11(a)	(description) sound gets quieter as distance increases or volume of sound decreases as distance increases (explanation) the amplitude of wave decreases as distance increases	2	each correct answer = 1 mark Note the answers must be comparative Accept ora Note sound gets quieter unqualified is not sufficient the answer must indicate somewhere that the distance increases. Note explanation must refer to information from the table Accept height of wave decreases Ignore comments about frequency, pitch and wavelength or amplitude is weaker Note if no other mark awarded accept correct relationship between amplitude and loudness
11(b)	any one from repeat increase the range of distances take more readings (within the range)	1	Accept increase the number of readings Accept Increase the distance Ignore changes to equipment Ignore references to accuracy and fair testing

Question	Answer	Marks	Further Information
12(a)	hydrochloric acid	1	Accept HCl, but name takes precedence Ignore hydrogen chloride
12(b)	M1 filtration or filter or description of filtration M2 collects filtrate	2	each correct answer = 1 mark Ignore decanting or sieving Accept (idea that) zinc carbonate is the residue on the filter paper
12(c)	leaves in a warm place or leaves by a window or slowly evaporates	1	Accept heat/boil to reduce volume by half and then leave (to evaporate) or heat to get saturated solution and then leave (to cool) or leave for a few days (to evaporate or crystallise) or leave solution to evaporate or heat and leave to cool Note crystallisation is not sufficient Ignore leave solution to dry/solidify Do not accept (idea of) heat to dryness